

Scott Haakenson

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EDUCATION

Michigan State University, College of Engineering

Bachelor of Science, Computer Science

GPA: 4.0/4.0 - Honors College

East Lansing, Michigan

May 2026

PROFESSIONAL EXPERIENCE

Rocket Companies, Enterprise Technology

Software Engineering Intern

Detroit, Michigan

May 2025 – August 2025

- Initiated and led the development of an Agile feature for the RHDS, the centralized service for all employee data, by building multiple endpoints in .NET Core, MySQL, AWS RDS, Kubernetes, and Docker
- Built an API that automated client access provisioning, replacing manual SQL updates; reduced change requests from 1.5 engineer-hours to minutes per request, saving ~52 hours annually
- Drove collaboration with product managers, a staff engineer, and internal stakeholders to clarify requirements and refine API contracts, ensuring usability, architectural alignment, and transparency throughout development

Michigan State University, CSE 331 – Data Structures and Algorithms

Undergraduate Learning Assistant

East Lansing, Michigan

January 2024 – May 2025

- Designed and implemented a Deques project with a 68-test automated grading suite in Python
- Evaluated 17 students' projects for time and space complexity, correctness, and adherence to best practices

Ford Motor Company, Model E Group

Product Development Intern

Dearborn, Michigan

May 2024 - August 2024

- Applied hyperparameter optimization to a physics-informed LSTM model that predicted high-voltage battery pack temperatures, improving the model's maximum error from 2.2°C to 1.6°C
- Designed backend scripts for comprehensive data analysis from various high-voltage battery tests in Python

Native Projex, Inc.

Backend Development Intern

Ho Chi Minh City, Vietnam

May 2023 - August 2023

- Enhanced search accuracy from 60% to 80% by automating the processing and categorization of unstructured data to engineer a refined dataset to train a fine-tuned version of Microsoft's DeBERTa machine learning model
- Developed the dynamic sort algorithm as an integral component of a filtering and sorting search framework to provide customers with the best matching locations within major cities in Southeast Asia in Python
- Implemented vector search using Weaviate to enable semantic search across 100,000+ locations

PROJECTS

Production-Calculator.com

October 2025 – Present

- Designed a scalable, three-tier full-stack system for modeling production pipelines, emphasizing stateless services and clean separation of concerns
- Implemented a three-layered .NET Core RESTful API and PostgreSQL with Flyway for DB schema migrations
- Built modern, secure authentication using JWT access tokens with opaque refresh tokens
- Developed a React/Next.js frontend with TanStack Query for client state management and Tailwind CSS
- Self-hosted infrastructure using Cloudflare Pages and Zero Trust, with deployments via GitHub Actions CI/CD

SKILLS

Programming Languages: C#, Python, C/C++, Java, SQL, GraphQL, TypeScript, HTML

Technical Proficiencies: .NET, Docker, Kubernetes, PostgreSQL, MySQL, AWS RDS, AWS S3, Flyway, REST APIs, GitHub Actions, React, Next.js, Tailwind CSS, Linux, Git